

Republic of the Philippines

Laguna State Polytechnic University

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ITEP 414 - System Administration and Maintenance

Enterprise Server Deployment - Installation Guide

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Portfolio Project: Junior System Administrator Deployment Documentation

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1. Project Overview

ABC Startup Solutions plans to deploy its first Linux server for future file sharing, remote administration, database hosting, web hosting, and internal services. As the Junior System Administrator, the task is to install Ubuntu Server, configure the required baseline settings, verify that the server works, and document the process so another administrator can reproduce the deployment.

This guide follows the supplied Week 3 module. Where the module requires values that can only come from the actual virtual machine - such as the DHCP-assigned IP address, exact partition scheme, filesystem, and screenshots - this document identifies the required evidence instead of inventing it.

2. Objectives

- Install Ubuntu Server in a virtual machine using the required enterprise baseline.
- Configure language, keyboard, networking, hostname, user account, disk, and OpenSSH.
- Verify login, hostname, IP address, Internet connectivity, updates, and SSH service status.
- Compare BIOS and UEFI and explain the Ubuntu boot sequence.
- Install Windows Server Evaluation for the bring-home activity and compare Windows Server, Ubuntu Server, and Rocky Linux.
- Produce professional documentation suitable for a GitHub portfolio.

3. Software Requirements

- Oracle VirtualBox or VMware Workstation.
- Ubuntu Server LTS ISO - latest stable version.
- Windows Server Evaluation ISO for the bring-home activity.

4. Virtual Machine Specifications

Component	Required value
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB virtual disk (VDI/VMDK)
Network	NAT, or Bridged if instructed
Optical Drive	Ubuntu Server ISO
Hostname	server01
Administrative user	gayoso - based on the module example "yourlastname"; verify against the actual VM
IP address	DHCP unless instructed otherwise; record actual assigned address

5. Step-by-Step Ubuntu Server Installation Procedure

5.1 Create the virtual machine

Create a new VM named Ubuntu-Server-Week03. Assign 4 GB RAM, 2 virtual processors, a 40 GB VDI/VMDK disk, NAT networking (or Bridged if instructed), and attach the Ubuntu Server ISO.

ACTUAL VM SCREENSHOT REQUIRED
Create the virtual machine evidence
Suggested file: screenshots/01-vm-settings.png

5.2 Language

Start the installer and select English.

ACTUAL VM SCREENSHOT REQUIRED
Language evidence
Suggested file: screenshots/02-language-keyboard.png

5.3 Keyboard Layout

Select the appropriate keyboard layout, such as English (US).

ACTUAL VM SCREENSHOT REQUIRED
Keyboard Layout evidence
Suggested file: screenshots/02-language-keyboard.png

5.4 Network Configuration

Accept DHCP unless instructed otherwise. Record the assigned IP address.

ACTUAL VM SCREENSHOT REQUIRED
Network Configuration evidence
Suggested file: screenshots/03-network-installer.png

5.5 Hostname and User Account

Set the hostname to server01. Create a non-root administrative user. This portfolio uses gayoso based on the module username example; the actual VM value should be verified. Use a strong password.

ACTUAL VM SCREENSHOT REQUIRED
Hostname and User Account evidence
Suggested file: screenshots/05-profile-hostname.png

5.6 Disk Partition

Choose Guided - Use Entire Disk. Record the actual partition scheme, filesystem used, and 40 GB disk size shown by the installer.

ACTUAL VM SCREENSHOT REQUIRED
Disk Partition evidence
Suggested file: screenshots/04-storage-layout.png

5.7 SSH Server

Enable Install OpenSSH Server.

ACTUAL VM SCREENSHOT REQUIRED
SSH Server evidence
Suggested file: screenshots/06-openssh-selection.png

5.8 Additional Packages

Do not install additional packages unless instructed.

ACTUAL VM SCREENSHOT REQUIRED
Additional Packages evidence
Suggested file: screenshots/06-openssh-selection.png

5.9 Complete Installation

Finish the installation, restart the virtual machine, and remove the installation ISO if prompted.

ACTUAL VM SCREENSHOT REQUIRED
Complete Installation evidence
Suggested file: screenshots/07-install-complete.png

6. Configuration Details

Setting	Required / recorded value
Language	English
Keyboard	Appropriate keyboard layout, e.g. English (US)
Network	DHCP unless instructed otherwise
Hostname	server01
Full name	Cedrick Gayoso
Username	gayoso - verify actual VM
Disk	40 GB - Guided - Use Entire Disk
Partition scheme	Record actual installer result
Filesystem	Record actual filesystem used
SSH	OpenSSH Server installed during setup
Additional packages	None unless instructed

7. Server Verification

After the first successful boot, perform all six verification tasks required by the module. Capture actual screenshots from the VM.

7.1 Login

Log in using the account created during installation.

ACTUAL VM SCREENSHOT REQUIRED
Successful login
Suggested file: screenshots/08-login.png

7.2 Verify Hostname

Run: **hostname**

Required hostname: server01.

ACTUAL VM SCREENSHOT REQUIRED
hostname showing server01
Suggested file: screenshots/09-hostname.png

7.3 Verify IP Address

Run: **ip addr**

Record the actual DHCP-assigned IPv4 address. The project PDF does not provide a fixed IP address.

ACTUAL VM SCREENSHOT REQUIRED
ip addr showing actual IP
Suggested file: screenshots/10-ip-address.png

7.4 Test Internet Connectivity

Run: **ping -c 4 google.com**

Capture the successful response.

ACTUAL VM SCREENSHOT REQUIRED
successful ping response
Suggested file: screenshots/11-ping-google.png

7.5 Update the Server

Run:

sudo apt update

sudo apt upgrade -y

ACTUAL VM SCREENSHOT REQUIRED
update and upgrade process
Suggested file: screenshots/12-apt-update-upgrade.png

7.6 Verify SSH Service

Run: **systemctl status ssh**

The service should display active (running).

ACTUAL VM SCREENSHOT REQUIRED
SSH active (running)
Suggested file: screenshots/13-ssh-status.png

8. BIOS vs UEFI Comparison

The module requires a comparison covering definition, boot process, maximum disk support, partition style, security features, speed, advantages, disadvantages, and modern usage. See BIOS_vs_UEFI.pdf for the complete table and explanation.

9. Ubuntu Boot Process Flowchart

Required sequence: Power On -> BIOS/UEFI Initialization -> Boot Device Detection -> Boot Loader (GRUB) -> Linux Kernel -> init/systemd -> Services Start -> Login Prompt.

10. Windows Server Installation Summary

The bring-home activity requires installing Windows Server Evaluation in a new virtual machine. Minimum tasks are: install the operating system, assign a computer name, create an Administrator password, log in successfully, and capture installation screenshots.

Item	Recorded value / status
Operating system	Windows Server Evaluation
Computer name	Not specified by the module - record the actual name assigned during installation
Administrator password	Created during installation; do not publish the password
Login	Must be verified with an actual screenshot
Screenshots	Required from the actual Windows Server installation

ACTUAL VM SCREENSHOT REQUIRED
Windows Server installation / login evidence
Suggested file: screenshots/14-windows-server-installation.png

11. Operating System Comparison

The module requires comparison of Windows Server, Ubuntu Server, and Rocky Linux using licensing, user interface, package management, security, performance, typical enterprise use cases, advantages, and disadvantages. See OS_Comparison.pdf for the full table and discussion.

12. Problems Encountered and Solutions Implemented

Record only problems that actually occurred in the laboratory. No VM logs or screenshots were supplied with the assignment PDF, so this package does not fabricate problems or solutions. Use the format below after completing the lab.

Problem encountered	Solution implemented
Actual issue 1	Actual troubleshooting action and result
Actual issue 2	Actual troubleshooting action and result

13. Lessons Learned

The deployment process shows that a server installation should be treated as a controlled and documented procedure. The required baseline settings create a repeatable starting point, while the verification commands provide evidence that identity, networking, updates, and remote administration are working. The BIOS/UEFI and boot-process activities also provide a structured way to understand where startup failures can occur. Finally, the comparison of Windows Server, Ubuntu Server, and Rocky Linux demonstrates that operating-system selection should be based on workload, administration, security, performance, and enterprise requirements.

14. References

- Official Ubuntu Server documentation for implementation details.
- Official Microsoft Windows Server documentation for the bring-home activity.
- Official Rocky Linux documentation for operating-system comparison research.
- UEFI Forum documentation/specifications for firmware research.